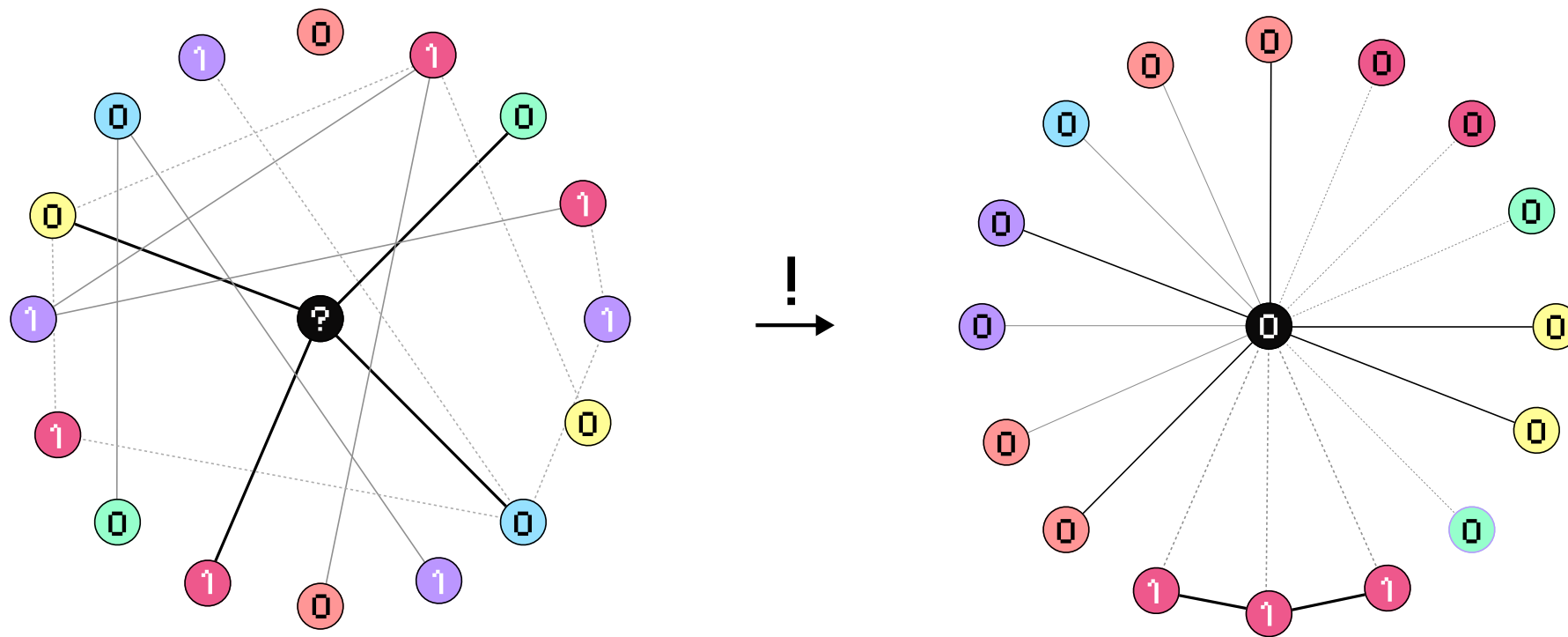


Communication Heterogeneity & Collective Consensus in Neural Cellular Automata

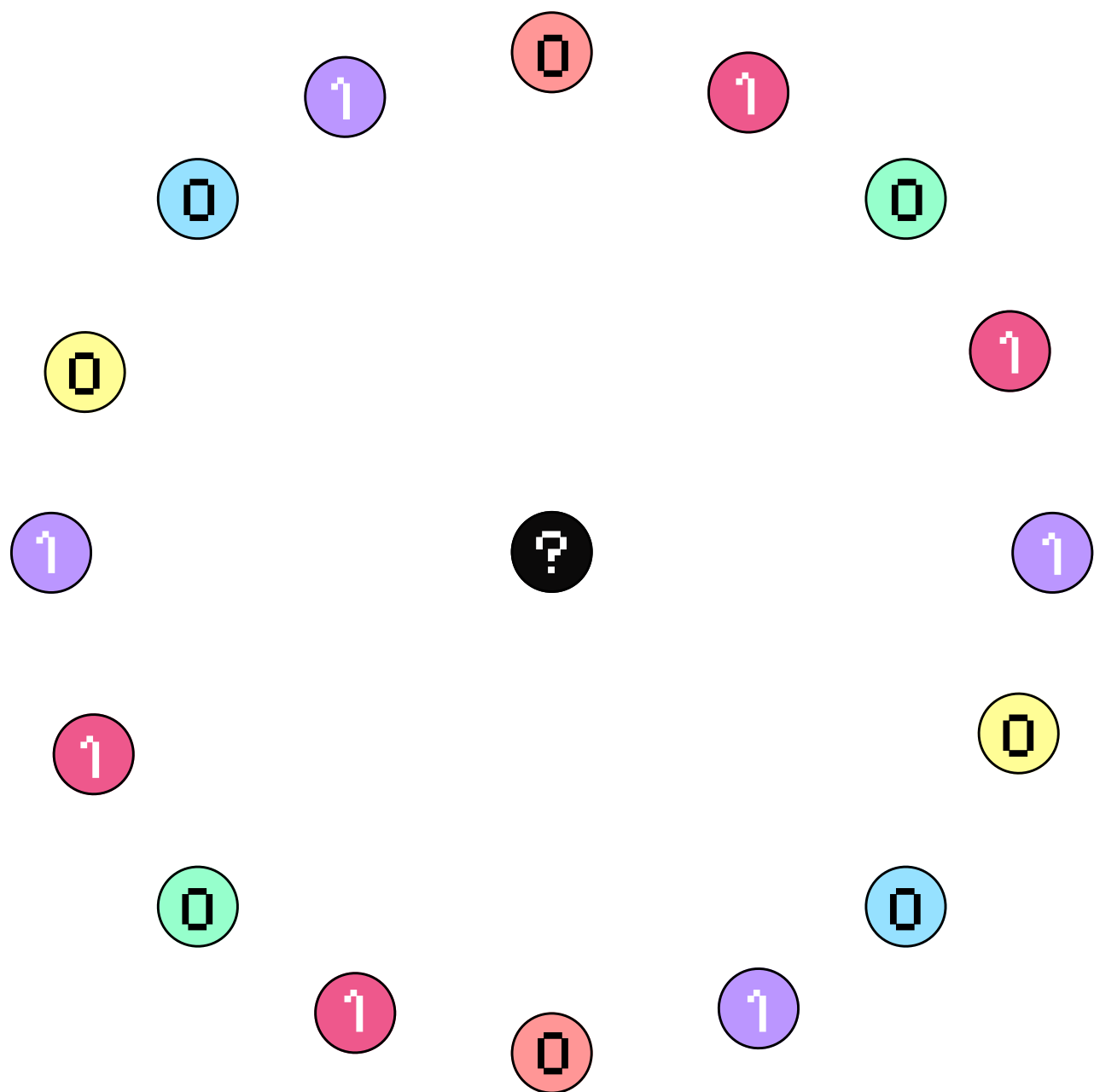


Nishit Singh

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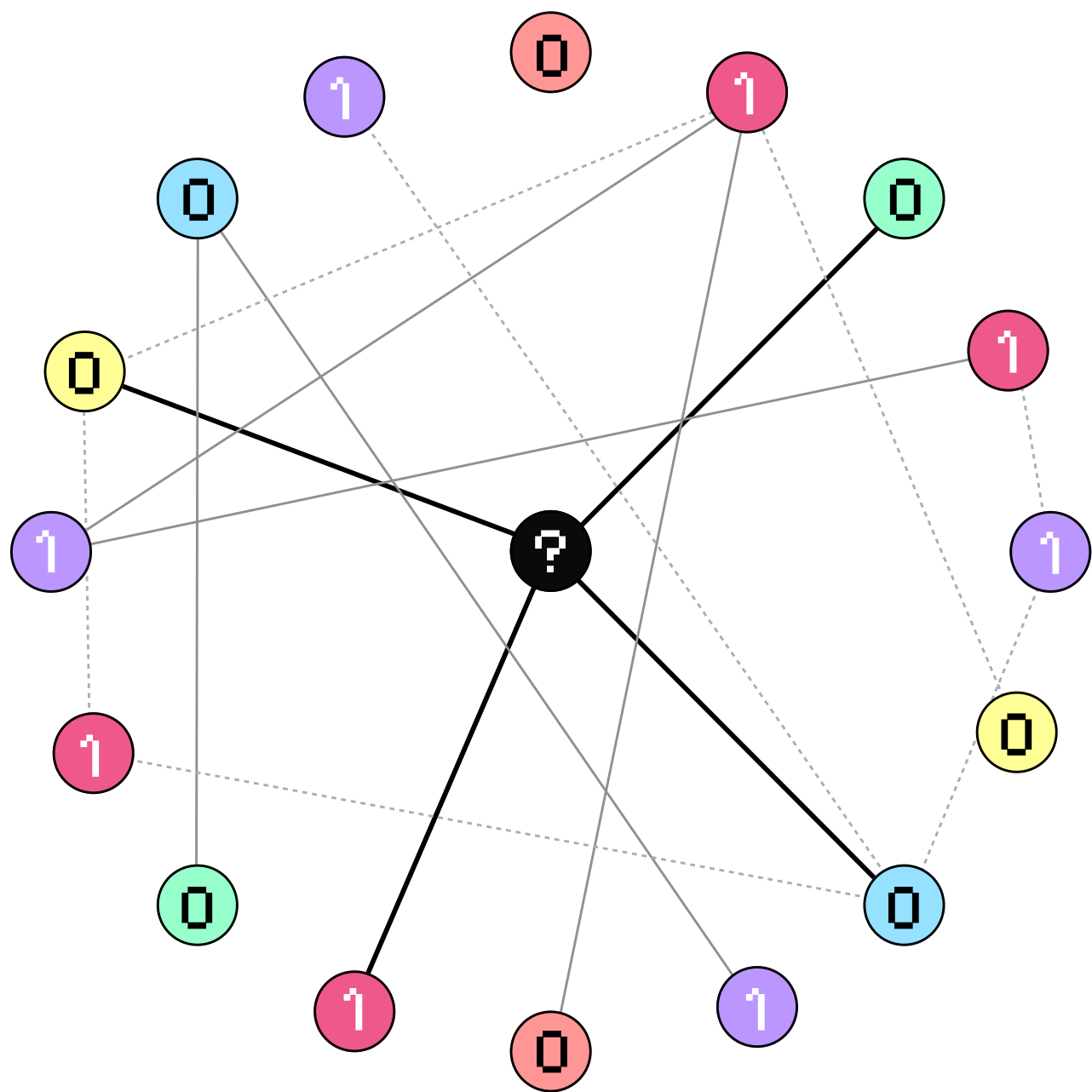
Density Classification

Density Classification



⬛ is the majority value in the initial configuration

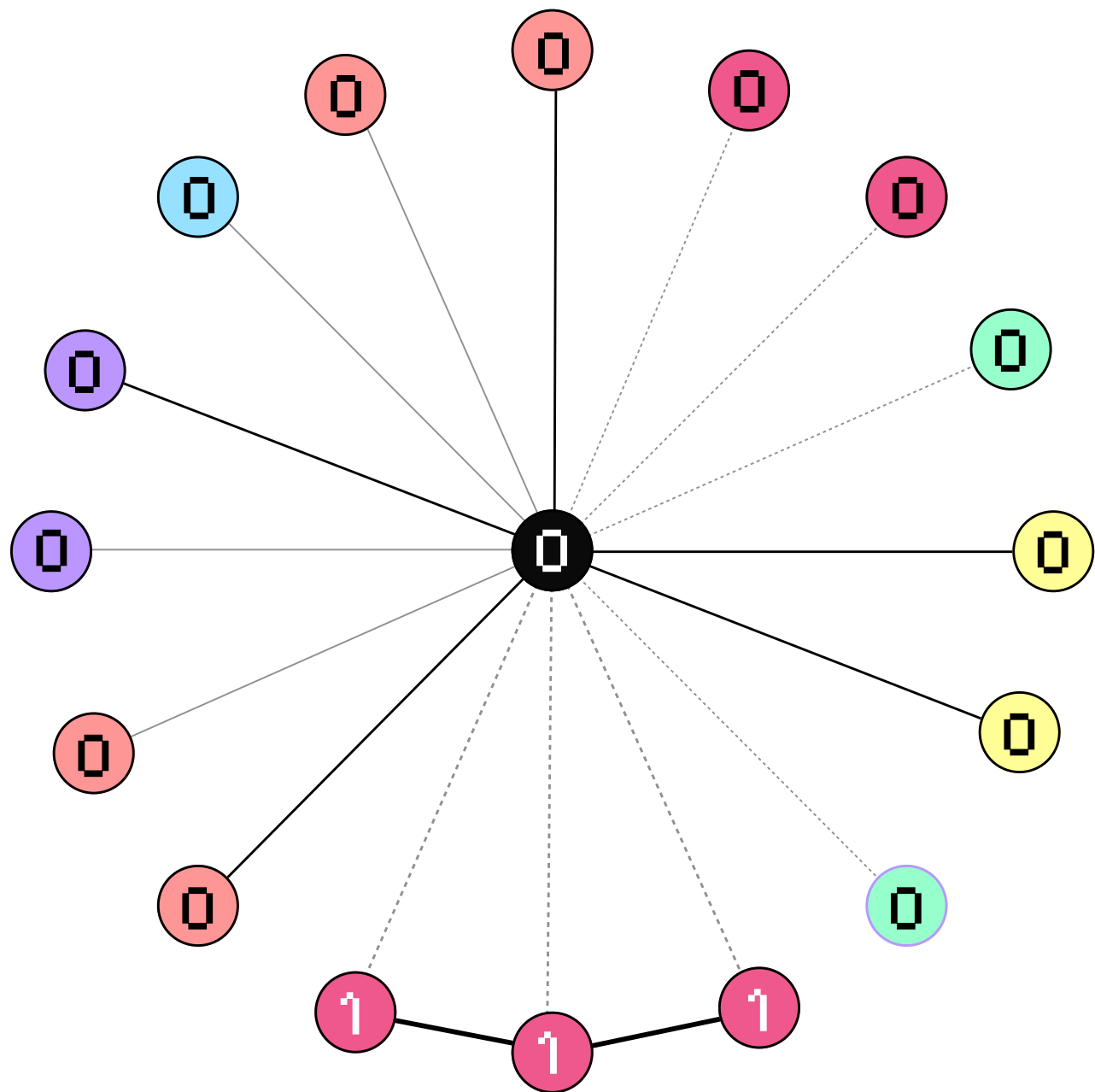
Density Classification



● is the majority value in the initial configuration

Only the nodes with edges between them can communicate

Density Classification

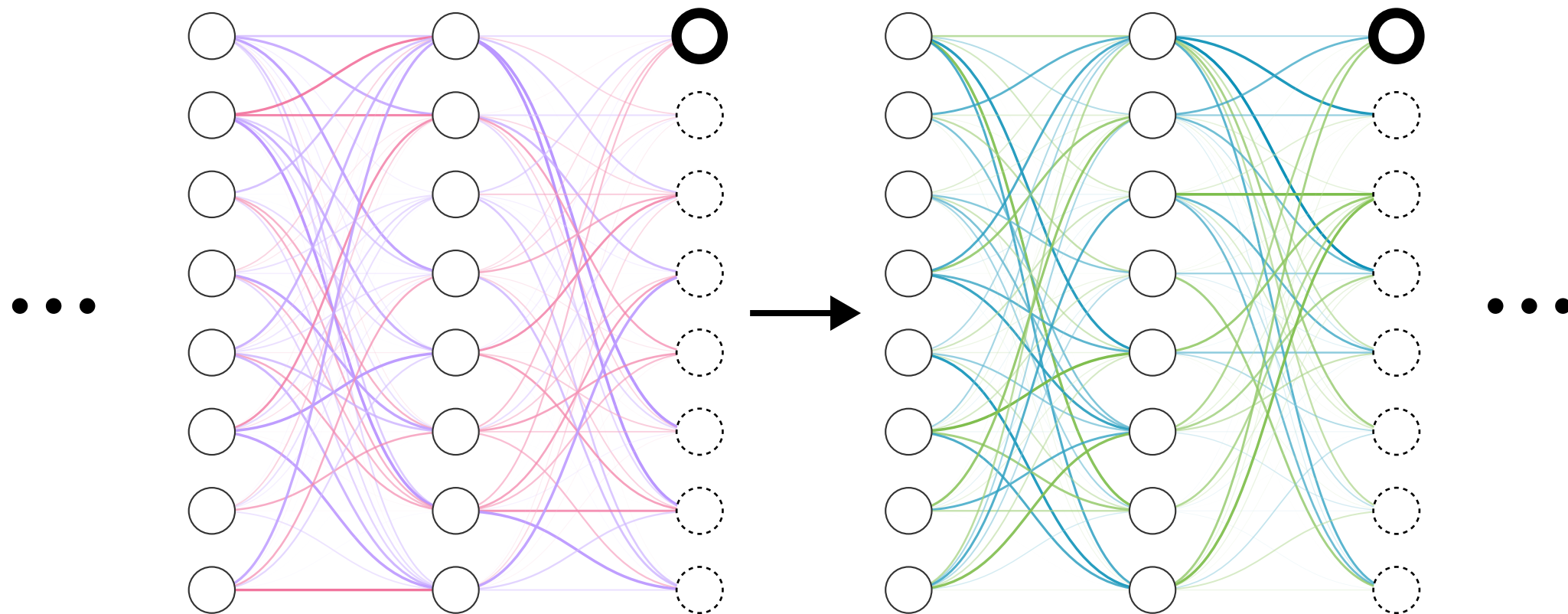


0 is the majority value in the initial configuration

Only the nodes with edges between them can communicate

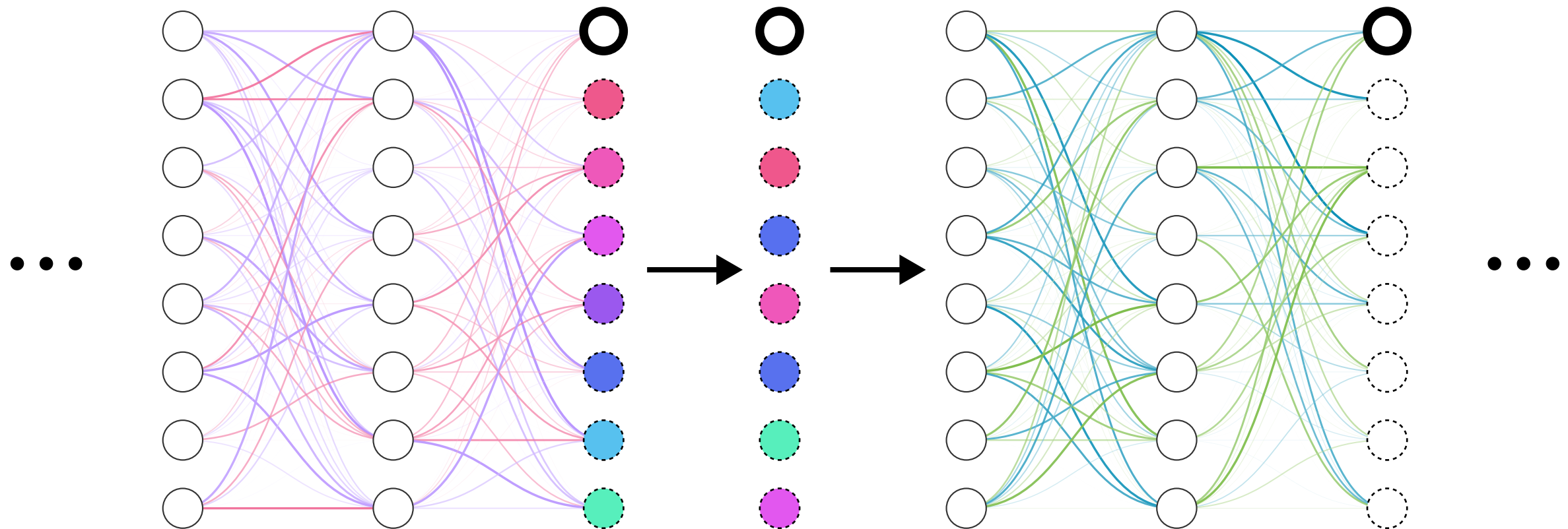
!!

Neural Cellular Automatas



Each network observes the vote^o and the (hidden + learnt) co-ordination signals^{*} from neighbours

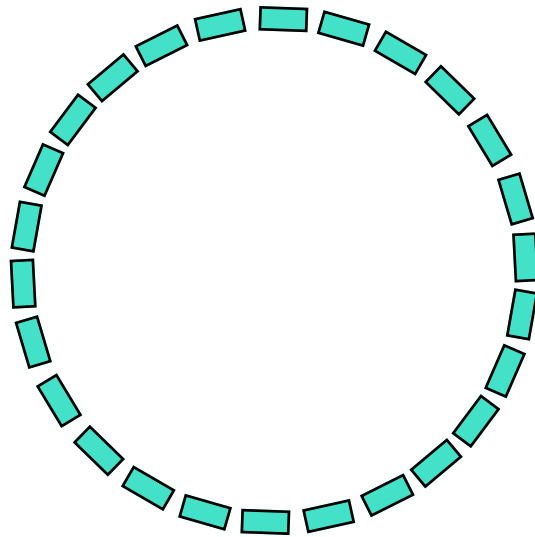
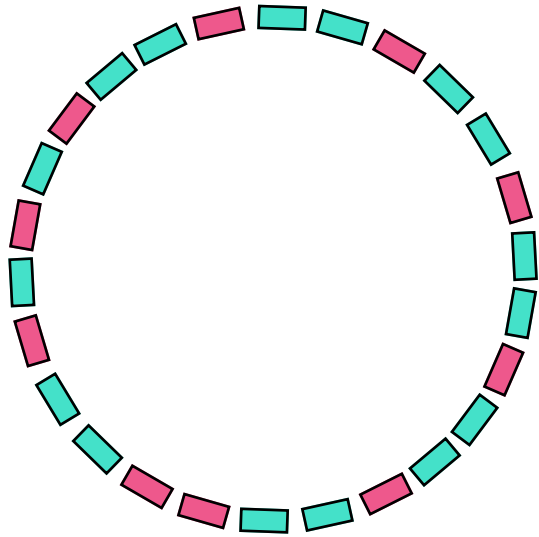
Neural Cellular Automatas



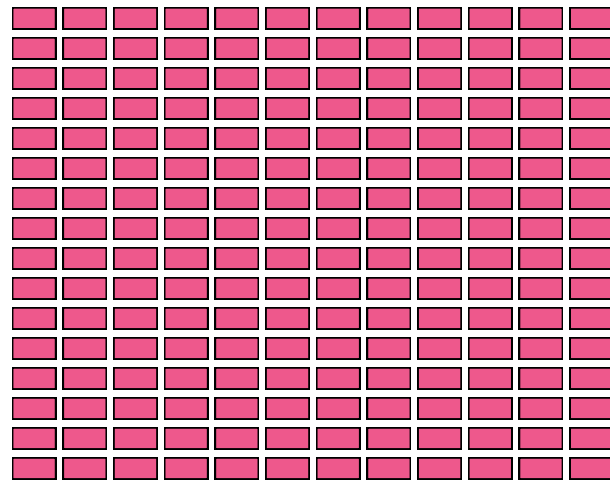
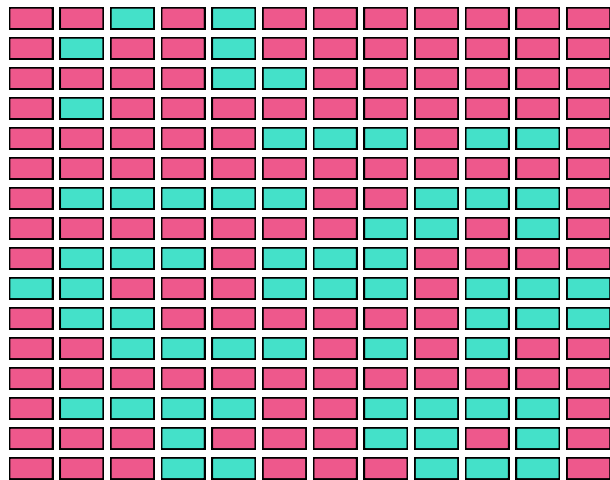
We can *permute* the channels by shuffling them!

$$\mathbf{T}(d) = (1-d)\mathbf{I} + (d)\mathbf{M}$$

Topologies

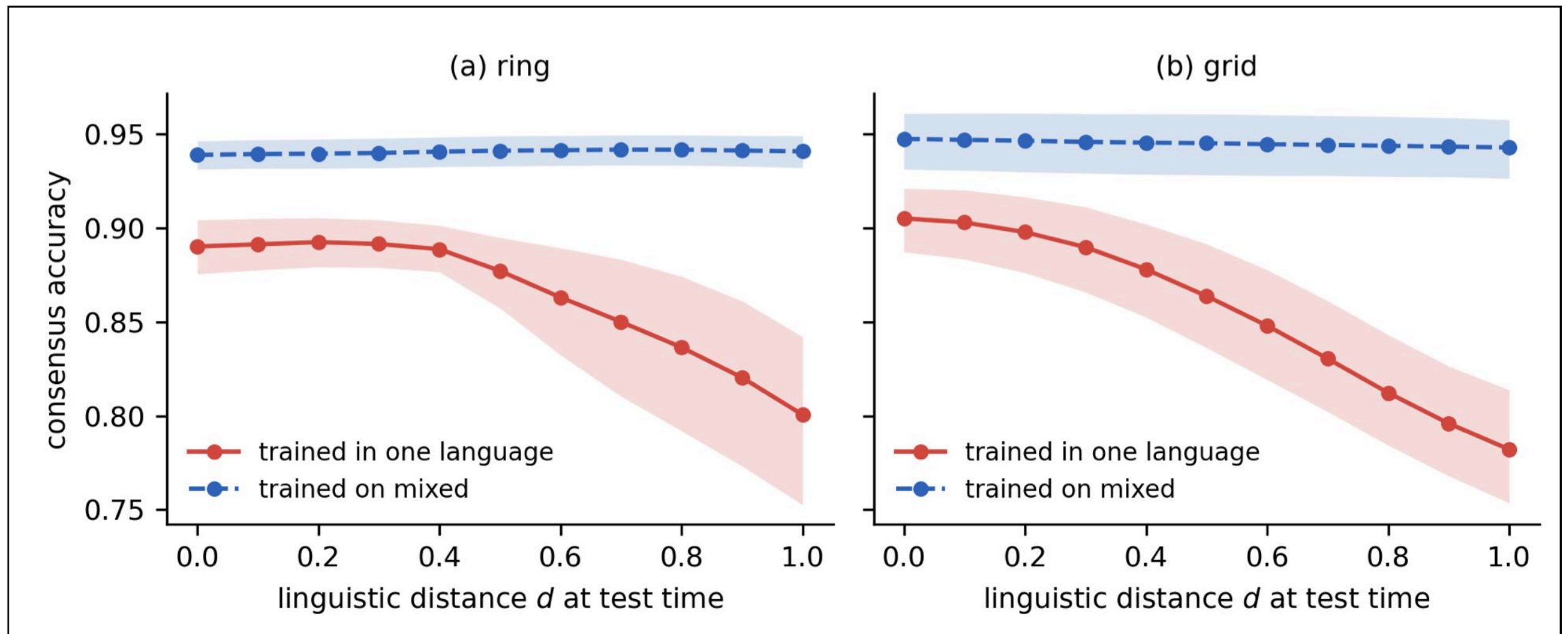


Only communicate with
neighbours immediately
left-right



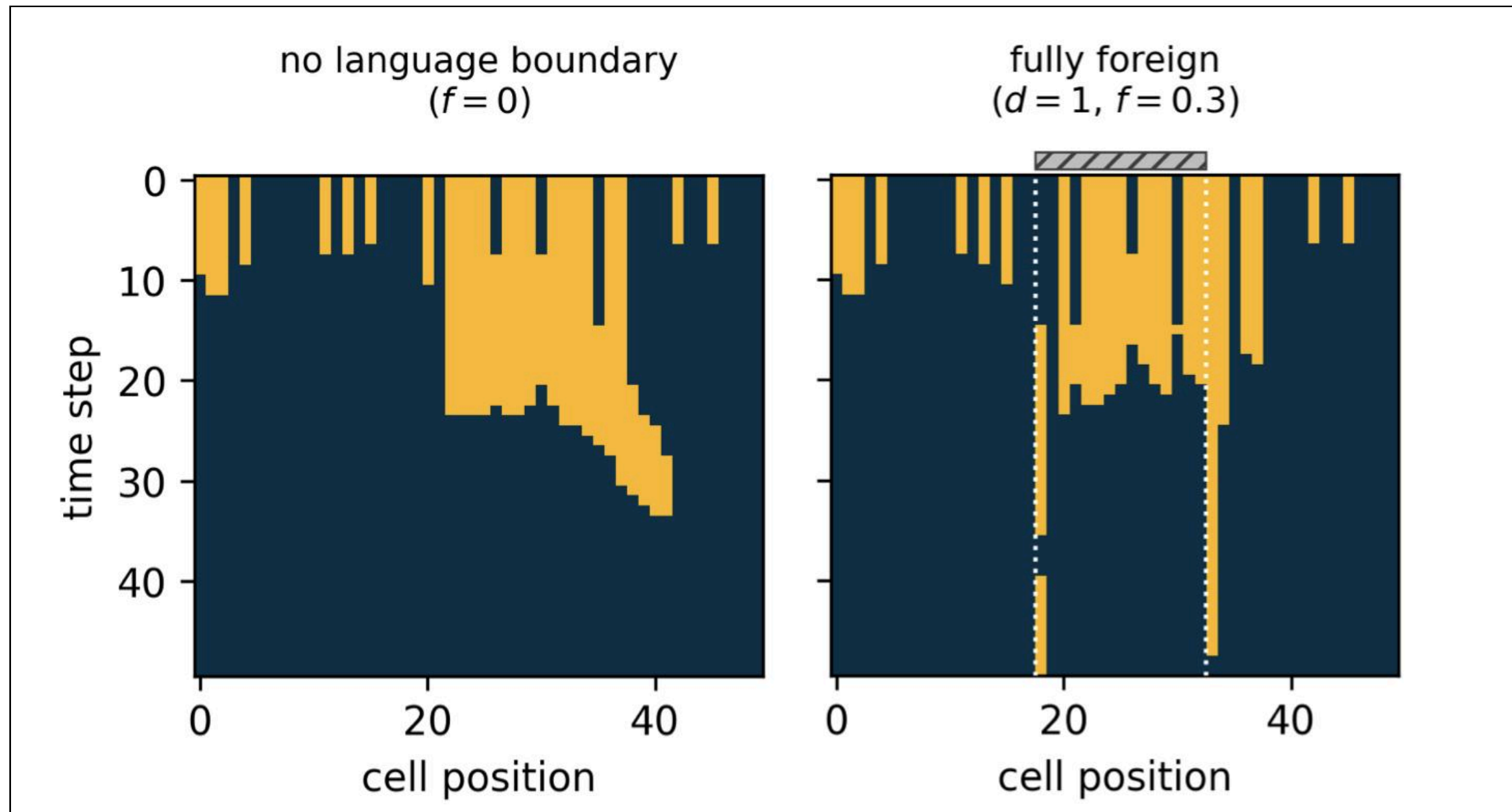
Only communicate with
neighbours immediately
left-right-front-back

Results



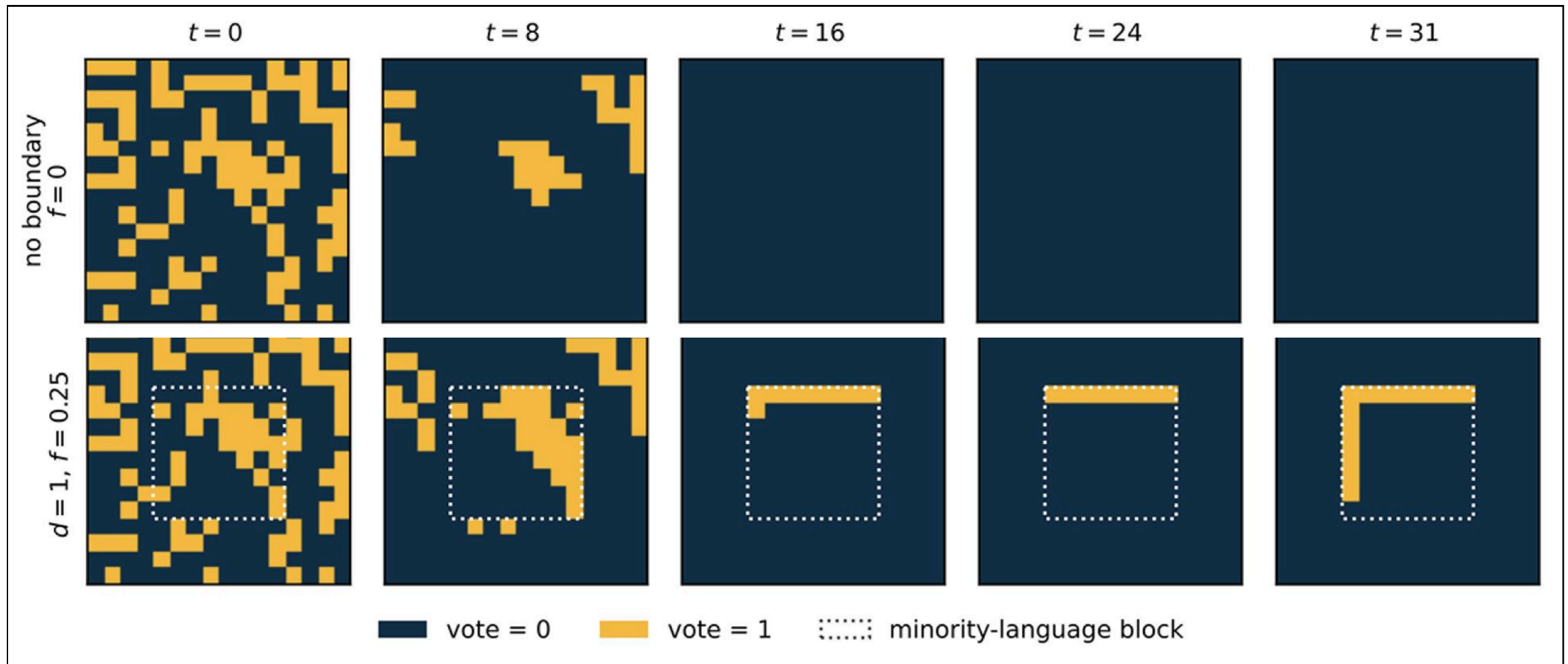
When you train on multiple “languages”, accuracy of consensus increases!

Results



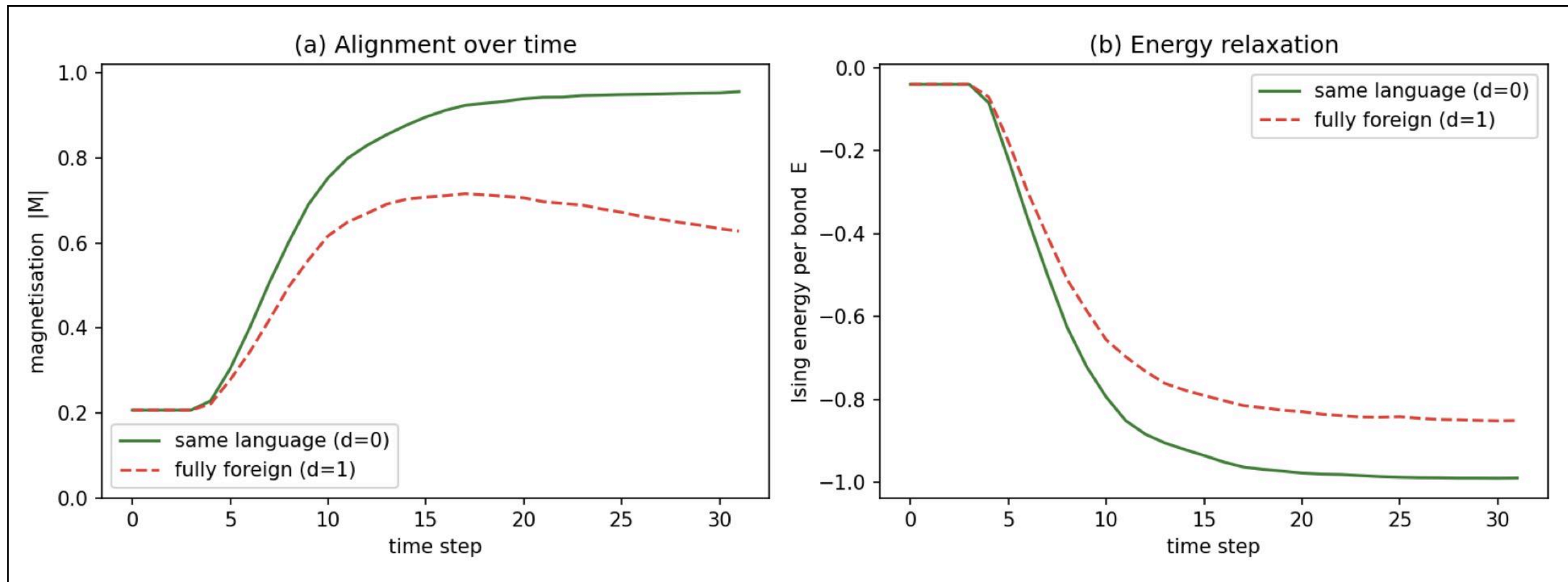
(RING) Inaccuracies usually live at the interface.

Results



(GRID) Inaccuracies usually live at the interface.

Results



$$\sigma_i = 2\hat{y}_i - 1 \in \{-1, +1\}$$

$$M = \frac{1}{N} \sum_i \sigma_i$$

$$E = -\langle \sigma_i \sigma_j \rangle$$

Could be modeled as an
Ising relaxation!

ty :D